Named Discrete Distributions, Pt 2

CSU, Chico Math 350

2026-03-30

outline

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Concept

A random variable with a negative binomial distribution originates from a similar case to the geometric distribution. Again, we focus on independent and identical trials, each of which results in one of two outcomes, success or failure. The probability of success p stays constant for each trial.

The geometric case handles the number of cases until the first success occurs. What if we are interested in knowing the number of trials until the second, third, fourth, etc. success occurs?

* How many people do you have to meet at college before you meet the 4th person from your hometown?
* Tire Mart has a lot of really cheap tires, but 20% of them are defective. How many tires do you need to go through to find 4 working new tires?

Random variable: Let X be a random variable that denotes the number of failures *before* the nth success, with probability of success p.

Negative Binomial Properties

Distributional Notation

$X \sim NegBin(n, p)$

PMF

$P(X = x) = \binom{x+n-1}{x} p^n (1-p)^x; \qquad x = 0, 1, 2, \ldots$

Mean & Variance

$E(X) = \frac{n(1-p)}{p}$

$Var(X) = \frac{n(1-p)}{p^2}$

(NOTE: the sum of n geometric distributions)

R Commands:

`dnbinom(x,size=n,prob=p)` to compute $P(X == x)$

`pnbinom(x,size=n,prob=p)` to compute $P(X \leq x)$

`rnbinom(N,size=n,prob=p)` to randomly draw N samples from a $X \sim NegBin(n, p)$ distribution.

Example 1: Oil Discovery

A geological study indicates that an exploratory oil well drilled in a particular region should strike oil with probability 0.2. We are interested in when the third oil strike hits. Let X be the number of dry wells (no oil) before the third oil strike (oil found).

Our parameters are:

$$n = 3, p = 0.2.$$

The PMF is:

$$X \sim NegBin(3, 0.2)$$

$$P(X = x) = \binom{x+3-1}{x}(0.2)^3(0.8)^x; x = 0, 1, ...$$

Example 1a

Theoretical Approach:

$$E(X) = n * (1 - p)/p = (3 * 0.8)/(0.2) = 12$$

$$Var(X) = n * (1 - p)/p^2 = (3 * 0.8)/(0.2^2) = 60$$

Simulated Approach:

```
x <- rnbinom(10000, 3, 0.2)
mean(x)
## [1] 11.9084
var(x)
## [1] 61.23253
```

Example 1b

Find the probability that the third oil strike comes on the **fifth** well drilled (i.e. find $P(X = 2)$).

By hand with PMF

$$P(X = 2) = \binom{4}{2}(0.2)^3(0.8)^2$$

```
choose(4,2)*(0.2^3)*(0.8^2)
## [1] 0.03072
```

Theoretical Approach with R:

```
dnbinom(2,3,0.2)
## [1] 0.03072
```

Simulated Approach:

```
x <- rnbinom(10000, 3, 0.2)
mean(x==2)
## [1] 0.03061
```

Exercise 1

Ten percent of the engines manufactured on an assembly line are defective. Answer the problems below related to finding non-defective engines:
* success = non-defective engine
* P(success) = p = 0.9
* Let X be the number of defective engines **before** the first non-defective engine is found.

1. What is the probability that the first non defective engine will be found on the second trial? (from above)

2. What is the probability that the third non defective engine will be found on the fifth trial?

3. What are $E(X)$ and $Var(X)$ for part 1?

4. What are $E(X)$ and $Var(X)$ for part 2?
Be sure you mention which approach you use (theoretical or simulated).

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Concept

A Poisson process is one where events occur at random times during a fixed time period. The events occur independently from each other, but with a constant average rate over that time period. Examples include:

* Number of calls per hour at a call center
* Number of hits on a webpage in a day
* Number of meteor strikes on the surface of the moon annually

Random variable: Let X be the number of events occurring in a Poisson process with rate λ over one unit of time (e.g. per year, per day, per second).

Poisson Properties

Distributional Notation
$X \sim Poisson(\lambda)$

PMF
$P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}$

Mean & Variance
$E(X) = Var(X) = \lambda$

R Commands:
`dpois(x,prob=lambda)` to compute $P(X == x)$
`ppois(x,prob=lambda)` to compute $P(X \leq x)$
`rpois(N,prob=lambda)` to randomly draw N samples from a $X \sim Poisson(p)$ distribution.

Example 2: Meteors

The Taurids meteor shower is visible on clear nights in the Fall and can have visible meteor rates around five per hour. What is the probability that a viewer will observe exactly eight meteors in two hours? Let X be the number of observed meteors in two hours:

$$X \sim Poisson(10)$$

$$P(X = x) = e^{-10} \frac{10^x}{x!}$$

Example 2: Meteors

Theoretical Approach By Hand:

$$P(X=8) = (e^{-10})\frac{10^8}{8!}$$

```
exp(-10)*10^8/factorial(8)
## [1] 0.112599
```

Theoretical Approach with R:

```
dpois(8,10)
## [1] 0.112599
```

Simulated Approach:

```
meteors <- rpois(10000, 10)
mean(meteors == 8)
## [1] 0.1184
```

Exercise 2

Suppose a typist makes typos at a rate of 3 typos per 10 pages. What is the probability that they will make at most one typo on a five page document?

Once again, you'll want to try both theoretical and simulated approaches.

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Concept

The hypergeometric distribution is a series of Bernoulli trials that are dependent. This occurs when we are *sampling without replacement* from a finite population. Examples include:

* Capture some fish, tag them, and release them. Then come back later and fish some more, counting how many tagged ones you catch again.
* Create a college student committee of 5 members, and count the number of graduate students chosen.
* Consider a population of voters, 40% of which favored candidate Jones. Ten voters are selected at random, and the number of voters favoring Jones is observed.

Random variable: Let X be the number of successes out of a sample size of k when drawing without replacement from a pool where there are a total of m successes and n failures available.

Hypergeometric Properties

Distributional Notation

$X \sim Hypergeom(m+n, n, k)$

PMF

$P(X = x) = \frac{\binom{m}{x}\binom{n}{k-x}}{\binom{m+n}{k}}$

Mean & Variance

$E(X) = k(\frac{m}{m+n})$
$Var(X) = k(\frac{m}{m+n})(\frac{n}{m+n})(\frac{m+n-k}{m+n-1})$

R Commands:

`dhyper(x, m, n, k)` to compute $P(X == x)$
`phyper(x, m, n, k)` to compute $P(X \leq x)$
`rhyper(N, m, n, k)` to randomly draw N samples from a $X \sim Hypergeom(m+n, n, k)$ distribution.

Example 3: Oh No! More Urns

An urn contains nine chips, five red and four white. Three are drawn out at random without replacement. Let X denote the number of red chips in the sample. Identify the parameters, and find $E(X)$ and $Var(X)$

$$X \sim Hypergeom(9, 4, 3)$$

$$m = 5 \text{ (red chips)}$$
$$n = 4 \text{ (non-red chips)}$$
$$k = 3 \text{ (sample size taken)}$$
$$total = m + n = 9 \text{ (total chips combined)}$$

Example 3: Oh No! More Urns

Theoretical Approach By Hand:

$$E(X) = k * (m/(total)) = 5/3$$
$$Var(X) = k * (m/total) * (n/total) * ((total - k)/(total - 1)) = 5/9$$

Simulated Approach:

```
n.red.chips <- rhyper(10000, 5, 4, 3)
mean(n.red.chips)
## [1] 1.6831
var(n.red.chips)
## [1] 0.5597304
```

Example 4: Gone Fishing

In a small pond there are 50 fish, 10 of which have been tagged. If a fisherman's catch consists of 7 fish, selected at random and without replacement, and X denotes the number of tagged fish, what is the probability that exactly 2 tagged fish are caught?

$$X \sim Hypergeom(50, 40, 7)$$

$$m = 10 \text{ (tagged fish)}$$
$$n = 40 \text{ (non-tagged fish)}$$
$$k = 7 \text{ (fisherman's catch / sample taken)}$$
$$total = m + n = 50 \text{ (total fish combined)}$$
$$\text{Find } P(X = 2)$$

Example 4: Gone Fishing

Theoretical Approach By Hand:

$$P(X = x) = \frac{\binom{10}{2}\binom{40}{5}}{\binom{50}{7}}$$

```
choose(10,2) * choose(40,5) / choose(50,7)
## [1] 0.2964463
```

Theoretical Approach with R:

```
dhyper(2, 10, 40, 7)
## [1] 0.2964463
```

Simulated Approach:

```
n.tagged.fish <- rhyper(10000, 10, 40, 7)
mean(n.tagged.fish == 2)
## [1] 0.2981
```

Exercise 3

Suppose that there are 3 defective items in a lot of 50 items. A sample of size 10 is taken at random and without replacement. Let X denote the number of defective items in the sample.

1. What is the probability that the sample contains **exactly** 1 defective item?

2. What is the probability that the sample contains **at most** 1 defective item? (Hint: R commands make this easier)

Exercise 4

A display case contains thirty-five diamonds, of which ten are real diamonds and twenty-five are fake diamonds. A burglar removes four gems at random, one at a time and (naturally) without replacement. What is the probability that the last gem she steals is the second real diamond in the set of four?

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references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Also adapted from notes by Dr. Roualdes, Dr. Donatello, and Dr. Gray.